

Executive Summary: Late Delivery Risk Prediction

Unified Mentor Project | Global Supply Chain Operations

1. Executive Overview

This project develops a machine-learning decision-support solution that estimates the probability that an order will be delivered late before shipment. The objective is to help operations teams identify orders requiring additional review and use available operational information proactively.

2. Scope and Data

The analysis uses 180,519 logistics orders with 40 original variables. The target contains 98,977 positive records (54.83%) and 81,542 negative records (45.17%).

3. Analytical Approach

The workflow included missing-value treatment, categorical encoding, numerical scaling, class-weighted classification, and operational feature engineering including Shipping Pressure Index, shipping-mode flags, Regional Congestion Indicator, and Order Complexity Score.

4. Model Results

Model	ROC-AUC	Precision	Recall	F1
Logistic Regression	0.7430	0.8463	0.5439	0.6622
Random Forest (initial)	0.8171	0.8483	0.6158	0.7136
XGBoost	0.7843	0.8362	0.5755	0.6817
Random Forest (cleaned/deployed)	0.7598	0.7947	0.6075	0.6886

The deployed model is the cleaned Random Forest. Identifier and location-proxy features were removed, and the deployed forest was reduced from 200 to 50 trees to reduce cloud resource requirements.

5. Operational Risk Framework

Low Risk: probability below 0.40. Medium Risk: 0.40–0.70. High Risk: above 0.70. These are initial operational thresholds and should be validated against actual intervention costs and service-level objectives.

6. Key Predictive Signals

Important predictive signals include profit and benefit measures, scheduled shipping days, order totals, sales per customer, discount measures, shipping-mode indicators, order complexity, regional congestion, and shipping pressure. Predictive importance does not establish causality.

7. Recommended Operational Use

- High-risk orders: prioritize proactive operational review.
- Medium-risk orders: monitor and review available capacity.
- Low-risk orders: continue standard processing and monitoring.
- Use region and shipping-mode analysis to support investigation.
- Combine model output with current logistics conditions and human judgment.

8. Dashboard

The solution is available through a live Streamlit dashboard with filters and order-level prediction functionality.

Live application: <https://late-delivery.streamlit.app/>

9. Governance and Limitations

The model is decision support, not an automatic decision-maker. Thresholds require operational validation, and performance should be monitored after deployment for drift and changing logistics conditions.

10. Conclusion

The project provides a complete prototype for pre-shipment late-delivery risk assessment, connecting machine-learning prediction with an operational dashboard while leaving final operational decisions with responsible teams.